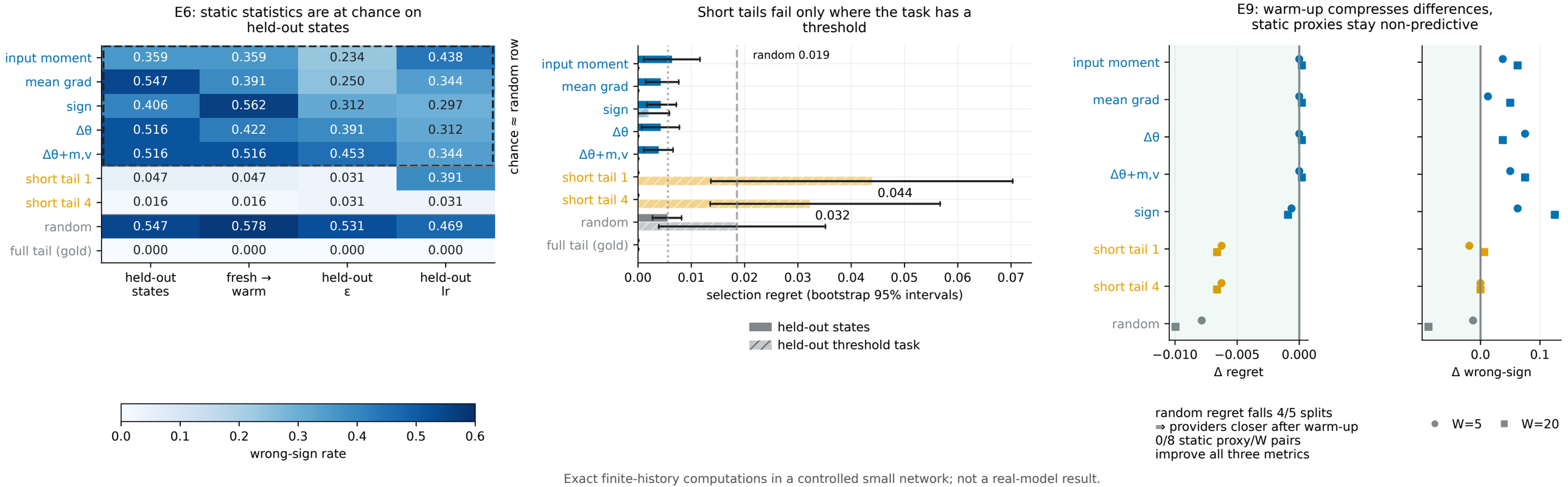
E6: static statistics are at chance on held-out states
input moment
mean grad
sign
Δθ
Δθ+m,v
short tail 1
short tail 4
random
full tail (gold)
0.359 0.359 0.234 0.438
0.547 0.391 0.250 0.344
0.406 0.562 0.312 0.297
0.516 0.422 0.391 0.312
0.516 0.516 0.453 0.344
0.047 0.047 0.031 0.391
0.016 0.016 0.031 0.031
0.547 0.578 0.531 0.469
0.000 0.000 0.000 0.000
held-out states
fresh → warm
held-out ε
held-out lr
0.0 0.1 0.2 0.3 0.4 0.5 0.6
wrong-sign rate
chance ≈ random row
Short tails fail only where the task has a threshold
random 0.019
0.044
0.032
0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07
selection regret (bootstrap 95% intervals)
held-out states
held-out threshold task
E9: warm-up compresses differences, static proxies stay non-predictive
−0.010 −0.005 0.000
Δ regret
0.0 0.1
Δ wrong-sign
random regret falls 4/5 splits
⇒ providers closer after warm-up
0/8 static proxy/W pairs
improve all three metrics
W=5
W=20
Exact finite-history computations in a controlled small network; not a real-model result.